14

Physiology of Retrograde Flow

There is nothing intuitive about retrograde flow.

You are pumping blood in one direction into the arterial system, which is the opposite direction that blood is supposed to flow. At first glance, this seems that it wouldn't make sense. Doesn't the aorta just become engorged? Does blood continue flowing backwards through the heart/lungs into the venous system? What happens to the tissues that are distal to the cannula (Fig. 14.1)?

Overall, retrograde flow involves understanding the overall directions of blood flow from the heart, the extracorporeal membrane oxygenation (ECMO) circuit, and throughout the vascular system. In this chapter, we will explore these as well as the potential effects that these can have on the body.

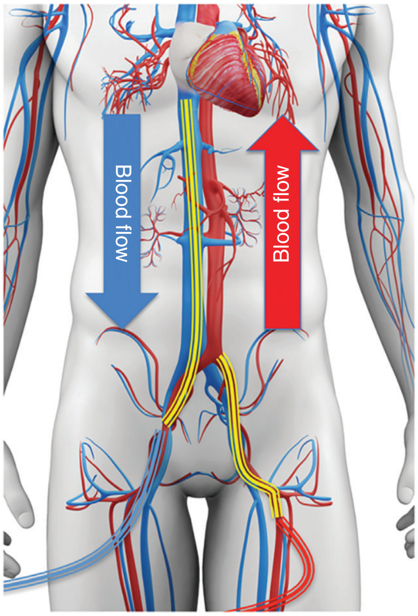

FIG. 14.1 Retrograde blood flow in VA ECMO. (Modified from SciePro/Shutterstock.com.)

THE DIRECTIONAL FLOW OF THE VASCULAR SYSTEM

If the network of blood vessels in the body was just a static system, then many of the above questions would hold true. However, as we have alluded to, the arterial/capillary/venous network functions as a very dynamic system that functions to drive blood flow forward, facilitating the forward/antegrade flow of blood.

How is this accomplished? The muscular arteries/arterioles contract along with the heart to drive blood forward, and the valved system of the veins accommodates this forward flow to eliminate pooling of blood, which may cause back pressure.

The overall effect is a high-pressure system that exists in the arterial circulation and a lower pressure system that exists in the venous system. This pressure differential drives blood forward and is responsible for filling pressure to the right side of the heart.

This vascular phenomenon is essential to answering our initial questions on blood flow during VA ECMO. What prevents blood from just engorging the aorta or continuing to flow backwards? How does any blood make it to the venous system where it can get drained by the venous cannula? The answer lies in this antegrade flow of the vascular system (Fig. 14.2).

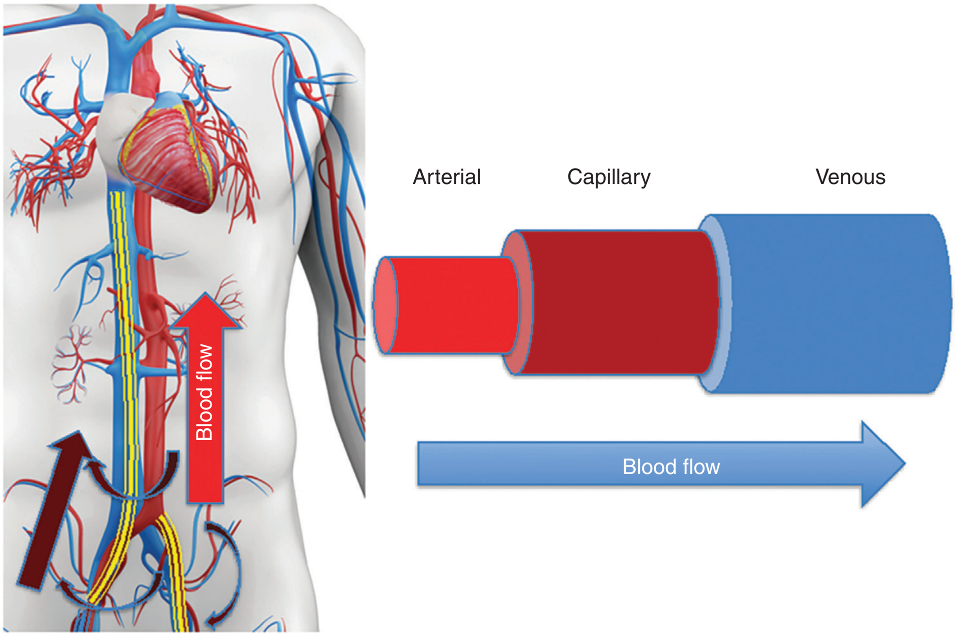

FIG. 14.2 The antegrade flow of the vascular pressure head. (Modified from SciePro/Shutterstock.com.)

Even if the heart is not beating at all (a phenomenon not uncommon in veno-arterial [VA] ECMO) there will still be antegrade flow from the arterial system through the capillary system into the venous system that allows for venous return. This antegrade flow of the vascular system can feasibly be driven by the ECMO flow, as the flow of blood from the ECMO circuit into the arterial circulation can serve to increase arterial pressure, increasing this pressure differential between arterial and venous systems.

THE NECESSITY OF RETROGRADE FLOW

Retrograde flow is a necessary consequence of peripherally inserted veno-arterial [VA] ECMO in adults. The intuitive manner of providing arterial blood flow would be antegrade, in order to pump blood in the direction of the heart and vascular system. However, there are both anatomic and functional limitations to doing so.

Accessing the heart/aorta directly requires surgical exposure. The carotid artery is used in children but is prohibitively risky in adults. The axillary artery can accommodate antegrade flow but is small enough that access requires grafting rather than percutaneous cannulation. This leaves us with the femoral artery, which is large enough to accommodate a cannula in most cases and can be accessed percutaneously, but requires flow to be directed retrograde through the aorta.

Let's now consider the physiologic effects of this retrograde flow.

The Effect of Retrograde Flow on the Left Ventricle

One of the most important implications of this retrograde flow is the effect on the left ventricle. The effect can be summarized in one word: afterload.

Remember afterload is the pressure that the heart must push against. You will remember from our initial introduction to afterload that the left heart is fairly tolerant of afterload, especially compared to the right heart (Fig. 14.3).

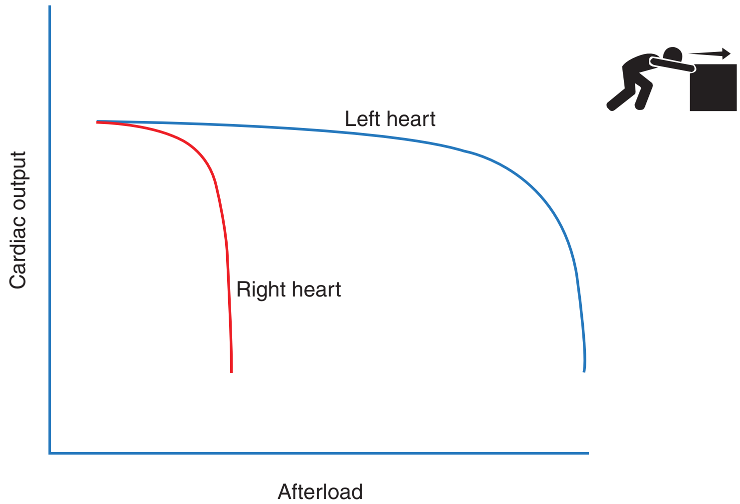

FIG. 14.3 Effects of afterload on the right and left heart. (Modified from Leremy/Stock Illustrations /Shutterstock.)

The primary reason for this tolerance is the function and anatomy of the left heart, which is more muscular and able to tolerate a higher pressure to work against. However, there is another phenomenon to consider, namely that as afterload increases, cardiac output drops. This in turn decreases mean arterial pressure (MAP), which decreases afterload.

In VA ECMO, this phenomenon is not necessarily the case, where the ECMO blood flow/afterload remains constant if the native cardiac output drops. This may mean that the effect of afterload due to retrograde flow will be more profound than would be normally observed if ECMO was not in place. The impact on the heart might be minimal or could lead to complete left ventricular dysfunction.

LEFT VENTRICULAR AFTERLOAD SENSITIVITY

Up until this point, we have generally described the afterload sensitivity of the left ventricle compared to the right ventricle, but there can also be a range of afterload sensitivity of the left ventricle depending on the clinical scenario. Let's consider the afterload effect of retrograde flow due to VA ECMO in two cases of cardiogenic shock.

Patient 1: Pulmonary embolism with preserved left ventricular function

Patient 2: Left anterior descending lesion with ischemic left ventricle

If both patients were placed on VA ECMO for cardiogenic shock, the responsiveness of their left ventricles may be very different. Patient 1 would be more tolerant to retrograde flow, with minimal impact on left ventricular output, while Patient 2 will experience a significant drop-off in left ventricular function/output (Fig. 14.4).

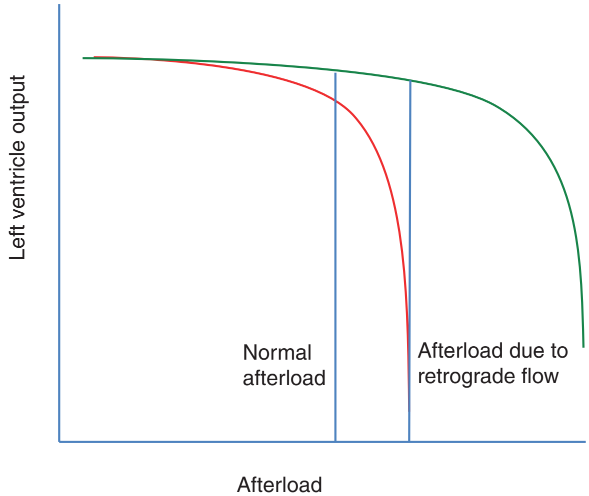

FIG. 14.4 The effect of additional afterload due to retrograde flow in a patient with primary right heart failure (green) versus primary left heart failure (red)

The degree of afterload sensitivity of the left ventricle will ultimately determine how it will be impacted by retrograde flow. In a case like Patient 1, there may be little to no effect of retrograde flow on the left ventricle, as the cardiogenic shock is only due to the failing right ventricle. In contrast, Patient 2 may experience a host of adverse effects to include worsening of the ischemia that caused the shock in the first place.

Let's further review these adverse effects of retrograde flow.

Adverse Effects of Retrograde Flow on the Left Ventricle

Imagine the case of Patient 2, our patient with ischemic cardiomyopathy and cardiogenic shock placed on VA ECMO. We gradually begin increasing the blood flow and notice a consequent increase in the MAP. But as MAP increases, you look at the arterial line tracing and notice a gradual narrowing, with a smaller and smaller difference in the pulse pressure. Continuing to increase the blood flow, the arterial line becomes flat. What is happening here?

As we have established to this point, the afterload has increased to the point that it has overwhelmed the ability of the left ventricle to eject blood. It may be still contracting (barely) but these movements may be insufficient to overcome the afterload from the ECMO circuit and the aortic valve may not actually open.

Thrombosis

If the aortic valve is not opening or is barely opening, this increases the static blood that exists around the valve and in the left ventricle. Even if anticoagulation is administered, the risk of clot formation is increased, a dangerous possibility in the arterial circulation due to the risk of valvular damage, stroke, or embolization.

Increased Left Ventricular Distension

As pulse pressure decreases and less blood is ejected out of the left ventricle, the ventricle can continue filling and become dilated.

How Do You Get Left Ventricular Distension If You Are on Complete ECMO Support?

If ECMO is draining all of the blood from the right ventricle and returning it to the arterial circulation, shouldn't there be no blood that is returning to the left ventricle? Unfortunately, this is not the case, even if the ECMO circuit is completely bypassing the heart. This is because there are blood vessels that drain directly into the left side of the heart as part of a normal, anatomic shunt. The cardiac veins drain into the Thebesian veins, some of which drain into the left side of the heart. Additionally, the bronchial circulation arises from the aorta, giving rise to the bronchial veins which drain directly into the left atrium. This means that even if your ECMO flow is greater than the cardiac output, the left ventricle will start to distend.

Increased Wall Stress on the Myocardium

Now you have a distended heart and increased afterload, both of which increase the amount of tension on the muscular ventricular wall. This increased wall tension decreases blood supply by compressing on the blood vessels supplying the myocardial cells of the heart. This can lead to worsening ischemia and myocardial damage, which can further any ischemia that precipitated the initial injury.

Elevated Pulmonary Pressures and Pulmonary Edema

As left ventricular pressures increase, it leads to increasing pressure in the left atrium, which eventually translates to increasing pressure in the pulmonary venous circulation. The result is higher hydrostatic pressure and accumulation of edema into the lungs. Pulmonary edema increases pulmonary shunt and will result in worsening oxygenation of any blood that is coming from the right side of the heart through the lungs. The effects of this hypoxic blood can be profound as we will discuss later (Fig. 14.5).

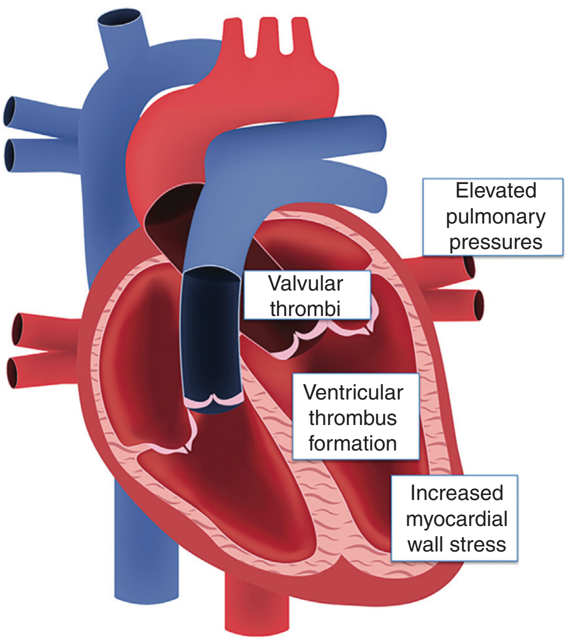

FIG. 14.5 Adverse effects of retrograde flow. (Modified from Ody_Stocker/Shutterstock.com.)

Venting the Left Ventricle

Because of the damaging effects of left ventricular distension, there exists a rationale for venting the left ventricle. This is an ability to allow blood to flow out of the left ventricle, improving distension, and preventing/mitigating the negative effects detailed earlier. Venting is not universal and varies based on practice patterns. Some pathology (for example, pulmonary embolism/primary right ventricular failure with preserved left ventricular function) may not need to be vented. However, once adverse effects of left ventricular distension such as thrombosis or pulmonary edema accumulate, it can be more difficult to manage, and the risk exists for rapid decompensation.

Potential options for venting include the following:

- **Chemical vent:** Use of inotropes to improve contractility and facilitate ejection of blood from the left ventricle.
- **Intra-aortic balloon pump:** Pump placed into the aorta via the femoral artery, facilitating the forward flow of blood through a counterpulsating balloon. Decreases afterload and improves coronary perfusion.
- **Percutaneous left ventricular assist device:** Motorized device that can facilitate the flow of blood from the left ventricle to the aorta distal to the aortic valve. Can be placed percutaneously from the contralateral femoral artery or surgically from the axillary artery.
- **Surgical drain:** Drainage of blood directly from the left ventricle, usually in the form of a drain from the apex of the left ventricle that is connected to the drain side of the circuit (see Fig. 14.6A). Usually, this is much more common in central cannulation as placement of this drain requires surgical access.
- **PA drain:** Rarely, if unable to drain from any other source, drainage of blood from the pulmonary artery can decompress the left side of the heart by draining out blood from the pulmonary circulation (see Fig. 14.6B). Cardiogenic shock in the setting of mitral stenosis may require this strategy, for example.

Left ventricular vents such as inotropes, intra-aortic balloon pumps, and percutaneous left ventricular assist devices have the advantage of being a bridge that ECMO can be weaned to as native recovery continues to progress.

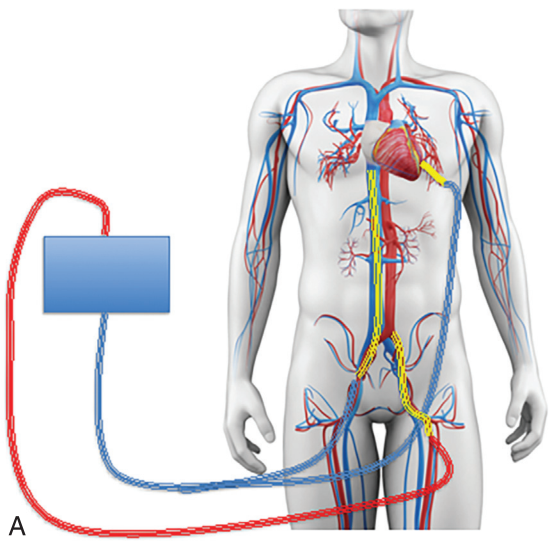

FIG. 14.6A Venting strategy: apical vent. (Modified from SciePro/Shutterstock.com.)

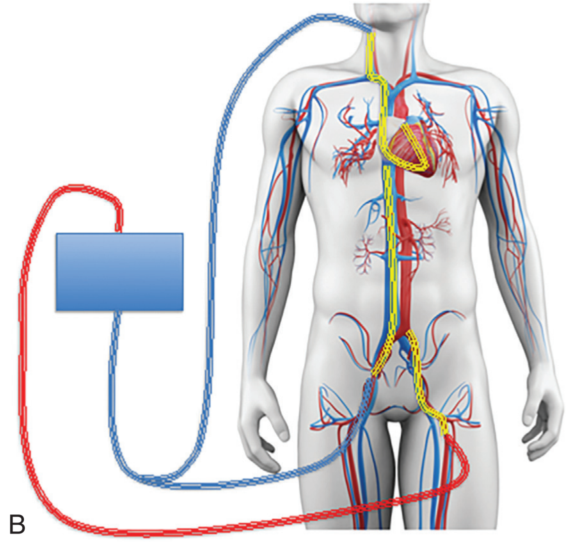

FIG. 14.6B Venting strategy: pulmonary artery (PA) drain. (Modified from SciePro/Shutterstock.com.)

WHICH WAY IS BLOOD FLOWING AGAIN?

One reason peripheral VA ECMO is confusing is we have to familiarize ourselves with competing directions of blood flow. We are accustomed to thinking of blood flow in one direction – out of the right heart, into the pulmonary circulation, into the left heart, out into the arterial circulation, across the vascular pressure head into the venous circulation, and into the right heart.

Now we have identified multiple competing directions of blood flow. Let's review in Fig. 14.7.

In the case of ECMO, we have drainage of venous blood into the ECMO circuit and return of oxygenated blood in a retrograde fashion as well as the maintenance of the vascular pressure head allowing for return of blood to the venous system and continued drainage through the circuit. We

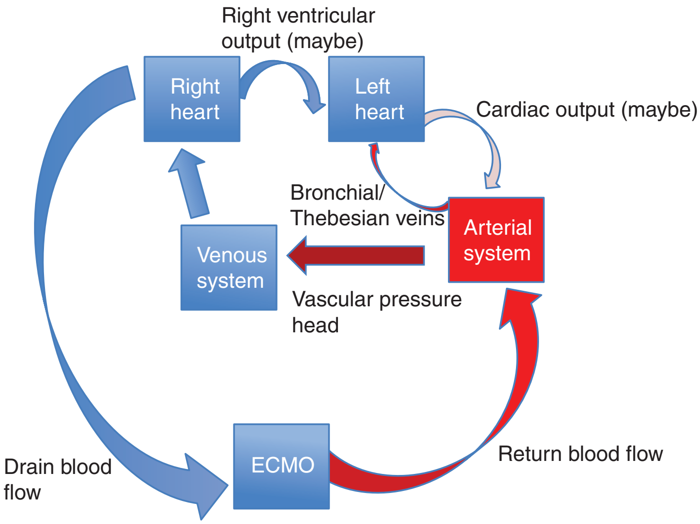

FIG. 14.7 Schematic of blood flow circulations on VA ECMO

also have drainage of bronchial/Thebesian veins into the left side of the heart. These circulations are maintained even if we are on complete ECMO support, with the heart doing nothing. On top of this, we can have antegrade blood flow from the right/left heart that can compete against the ECMO blood flow in the case of partial ECMO support or as the heart begins to recover. Let's now look at what this looks like clinically.

Competing Circulation on VA ECMO: Retrograde Versus Antegrade Blood Flow

The interplay between retrograde blood flow from the ECMO circuit and antegrade blood flow from the native heart/lungs is the fundamental dynamic to understand in peripheral VA ECMO. Understanding, identifying, and assessing these competing circulations will be essential to managing VA ECMO.

Let's return to our thought experiment, considering the physiology of our patient with ischemic cardiomyopathy. When we last left him, our flow was increased to the degree that there was no pulsatility on the arterial line, and the aortic valve was barely opening. (Fig. 14.8). Illustrates of what this would look like for the rest of the body.

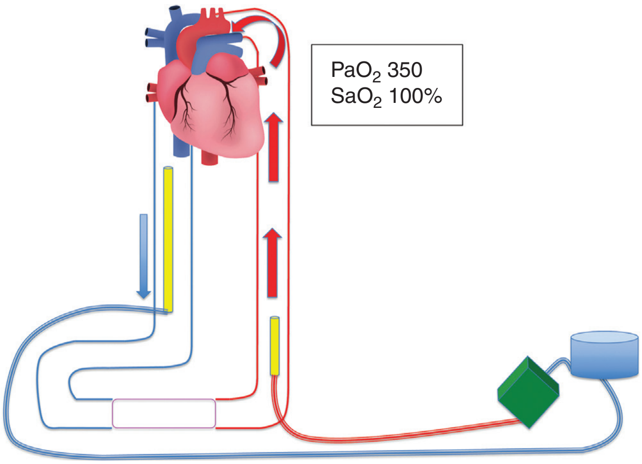

FIG. 14.8 VA ECMO oxygenation in patient with minimal native cardiac output. (Modified from Ody_Stocker/Shutterstock.com.)

As you can see, since there is no forward/antegrade flow from the heart, the oxygen content of blood returning from the ECMO circuit is exactly what the body sees. This includes the arterial circulation from the aorta all the way up to the coronary ostia. The oxygen delivery of the circuit translates so directly to the oxygen delivery of the body that we may consider decreasing FiO_2 to the ECMO circuit, so as not to cause cerebral damage from hyperoxia.

In this case, you have complete and efficient delivery of oxygen and all of the benefits of VA ECMO discussed in Chapter 13 (perfusion of abdominal organs, support of right heart, support of MAP, etc.). However, these benefits have to be weighed against the harmful effects of left ventricular distension/stasis.

You go home for the night, and return the next morning. To your pleasant surprise, you note that there is now pulsatility on the arterial line. An echocardiogram later in the morning shows an improvement in the left ventricular function with opening of the aortic valve. However, when you review the labs for the morning, you note that there is a PaO_2 of 60 mmHg on the blood gas where this had been 350 mmHg yesterday. What happened? Why did the oxygenation of the blood decrease?

In this case, we are seeing evidence of competing circulations (Fig. 14.9). While you had one circulation from the ECMO circuit that was unencumbered yesterday, today, there is clearly evidence that these two circulations are competing against each other. There is still blood draining from the arterial to the venous system via the vascular pressure head, however this is done via two competing flows, much like two hoses aimed into a drain will still drain in the same direction.

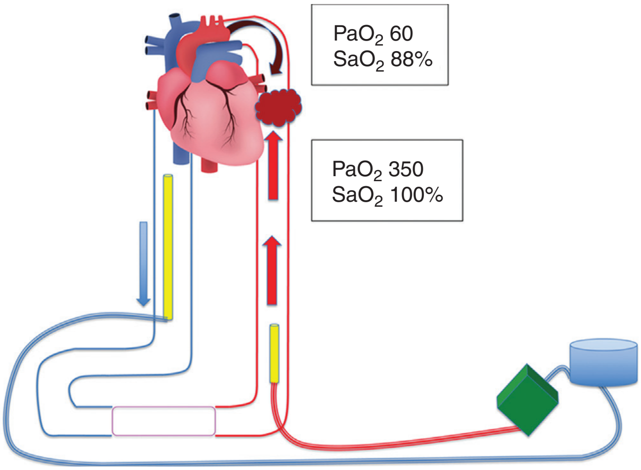

FIG. 14.9 VA ECMO oxygenation in patient with improving cardiac function. (Modified from Ody_Stocker/Shutterstock.com.)

How can we assess these two flows? Which blood flow is supplying what organ? Can we quantify this?

In general, ECMO blood flow should be assessed in the context of how perfusion is maintained. In other words, ECMO blood flow should be titrated based on the clinical indicators of perfusion: assessing whether we are maintaining an adequate MAP and urine output, reducing pressors, with lactate and liver function tests normalizing. If we are not maintaining these endpoints, then we need to address ECMO blood flow or adjust other support.

However, as competing antegrade flow from the heart/lungs increases, we will need to understand and quantify the effect of this circulation. We will do this by assessing the effect of this competing circulation on oxygenation.

Assessing Oxygenation of the Native Circulation: Role of the Right Radial Arterial Line

We now see that we have evidence of a mixing cloud somewhere between the heart and the ECMO return cannula, where blood proximal to the mixing cloud reflects the oxygenation of the native heart/lungs while blood distal to the mixing cloud reflects the oxygenation of the ECMO circuit.

Often teams will opt to place a right radial arterial line in patients with VA ECMO. Why is this? Looking at Fig. 14.10, as blood travels out of the heart, past the aortic valve, the first vessels supplied are the coronary arteries, the innominate artery, which gives rise to the right internal carotid artery, and the left internal carotid artery. Usually, this works in the body's favor having the coronary and cerebral blood vessels supplied first, prioritizing the oxygenation of these organ systems (Fig. 14.10).

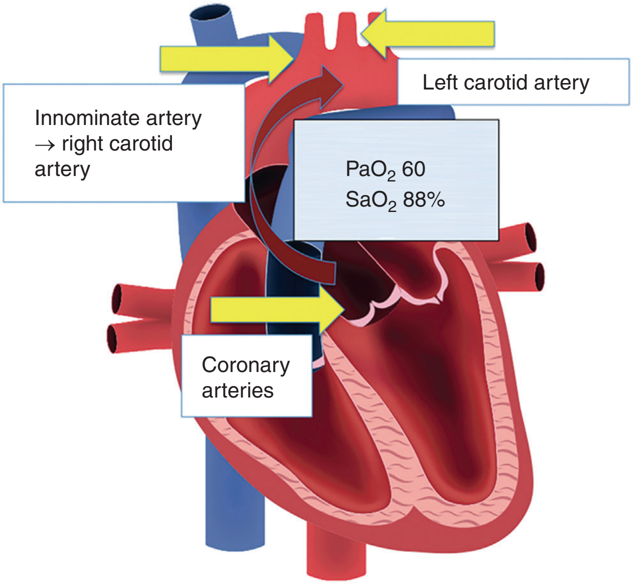

FIG. 14.10 Blood vessels initially supplied from blood leaving the left ventricle. (Modified from Ody_Stocker/Shutterstock.com.)

However, when we are on ECMO, and experiencing recovering cardiac function, this blood will reflect the oxygenation of whatever the lungs are able to supply (the saturation being 88% in Fig. 14.10).

The value of a right radial arterial line is that it reflects the oxygenation of what is being supplied to the innominate artery and will give the best sense of the circulation that is coming from the heart/lungs since it is the most proximal sample of the aortic supply that can be taken. If blood is oxygenated to the right radial artery, the assumption is that the same or greater oxygenated blood is being supplied to the cerebral circulation.

WORSENING HYPOXIA WHILE ON VA ECMO: IDENTIFICATION AND MITIGATION

Let's return to our patient. It is now ECMO day 3. The heart has been slowly recovering over the last several days. The chest X-ray has been gradually worsening, likely from an aspiration pneumonia that occurred during the initial arrest that triggered the cardiogenic shock.

We have been following blood gas results from a femoral line that was left in the contralateral groin as well as from a right radial arterial line that was placed and have noticed a gradual worsening of the oxygenation of the blood from the arterial line. The most recent blood gas results are as follows:

Right radial line ABG: 7.35, $PaCO_2$ 52, **PaO_2 60**, **SaO_2 75%**

Femoral line ABG: 7.49, $PaCO_2$ 30, **PaO_2 350**, **SaO_2 100%**

What explains the different results? As you can deduce, the femoral arterial line is reflecting the circulation from the ECMO circuit while the radial arterial line is reflecting the circulation from the heart and lungs.

This can be a common phenomenon to monitor on VA ECMO – upper body hypoxemia combined with lower body hyperoxia, sometimes referred to as harlequin phenomenon. The two components of this phenomenon are improving cardiac function and worsening pulmonary function (Fig. 14.11).

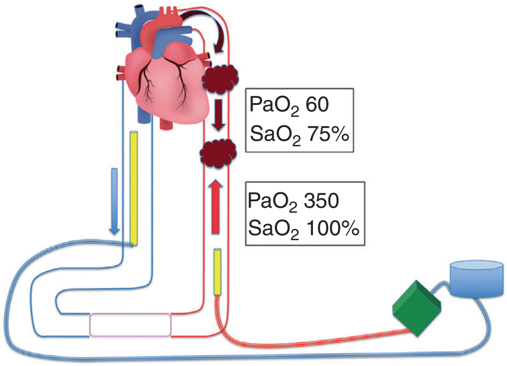

FIG. 14.11 VA ECMO oxygenation in patient with improving cardiac function and worsening lung function. (Modified from Ody_Stocker/Shutterstock.com.)

Why is this phenomenon common? It is a common pattern to see a drop in native cardiac output initially, in the initial phases of the injury, due to the effect of the afterload from the retrograde flow of the ECMO circuit, which can then slowly improve over the course of the next several days. However, it is also a common pattern to see the worsening of pulmonary function over the course of several days, either due to pneumonia as in this case or pulmonary edema due to insufficient left ventricular venting/volume overload/etc.

As a consequence, we must always maintain vigilance in monitoring for this phenomenon. If arterial blood gases are checked from a femoral arterial line, or even from a left-sided radial arterial line, the possibility exists of being falsely reassured of adequate oxygenation to the cerebral and coronary blood vessels, by sampling blood that is reflective of the ECMO circulation rather than the native heart/lungs. By checking for adequate oxygenation of the blood coming from the native heart/lungs, we are ensuring that oxygenated blood is being delivered to the coronary and cerebral vessels so as not to worsen any injury.

The second step is mitigation and correction of hypoxia once the phenomenon is recognized. If you recognize hypoxia in blood to the right radial arterial line, then we must improve oxygenation by optimizing the native heart and lungs. This can be done in a variety of ways to include:

- **Increase support from the ventilator**. Increasing PEEP or FiO_2 will usually improve oxygenation of blood being ejected from the native heart/lungs. Remember that this approach is counterproductive if you have to escalate support to the degree that it is injurious to the native heart/lungs.
- **Diurese/treat underlying cause**. If the cause is due to pulmonary edema, diuresis can help to fix the underlying pulmonary issues and will prevent from having to escalate support from the ventilator, as noted earlier. If due to other pulmonary etiologies (pneumonia, mucous plugging, bronchospasm) then attentive management of the underlying cause (timely antibiotics, secretion clearance, administration of bronchodilators) can be of benefit.
- **Address left ventricular venting strategy**. Intermittently, this hypoxia can be due to insufficient venting strategy. Mitigating strategies can include adding inotropes, adding another form of mechanical circulatory support (balloon pump, percutaneous left ventricular support device, etc.), or escalating to a more robust form of support (such as a surgically implanted temporary ventricular support device).
- **Transition ECMO configuration to Veno-arterial venous (VAV) ECMO**. If all else fails, and upper body hypoxia is refractory to any other interventions, then a transition to VAV

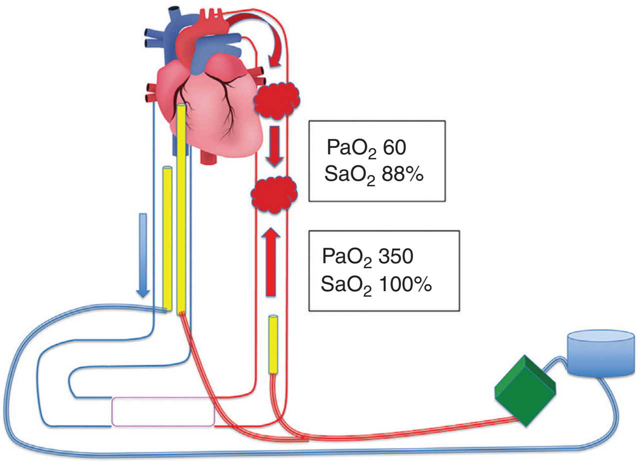

FIG. 14.12 VA ECMO oxygenation in patient on VAV ECMO. (Modified from Ody_Stocker/Shutterstock.com.)

ECMO may allow for circulatory support, while also providing oxygenated blood to the venous side of the circulation and improving hypoxia (illustrated in Fig. 14.12).

Let's examine this support configuration in a little more detail.

VAV ECMO in Supporting Upper Body Hypoxemia in Setting of Cardiogenic Shock

When we use VAV ECMO in this setting, we are adding a venous cannula either in the femoral vein or in the internal jugular/subclavian vein and connecting it to the return line of the ECMO circuit via a plastic Y connector (Fig. 14.13).

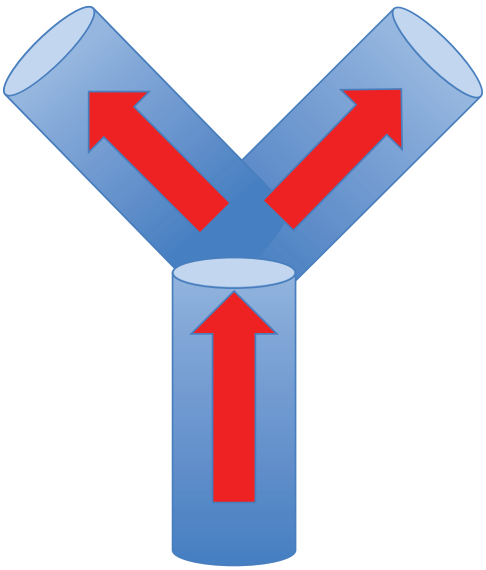

FIG. 14.13 Y connector dividing blood flow returned from ECMO circuit to two different cannulas

This connection divides oxygenated blood returning from the ECMO membrane/pump between the venous return cannula and the arterial cannula. Doing so allows for cardiac support and respiratory support simultaneously. The more blood flow that is directed toward the arterial cannula, the greater the cardiac support, while the more blood flow directed toward the venous return cannula, the greater the respiratory support.

TITRATION OF SUPPORT ON VAV ECMO

Let's say you transition our patient to VAV ECMO, in order to allow for better oxygenation. You cut in the Y connection and go back on flow. How would you imagine that flow is being divided between the arterial and venous limbs?

As you might have figured, based on our discussion of flow dynamics, the venous limb will have a lower resistance and, therefore, all else being equal, blood would flow preferentially into the venous limb (Fig. 14.14).

Therefore, on initiation of VAV ECMO, excess resistance will have to be placed on the venous limb, whether in the form of a partially occlusive pump clamp or an adjustable clamp. After this, then flow can be adjusted based on the desired support to the venous or the arterial side.

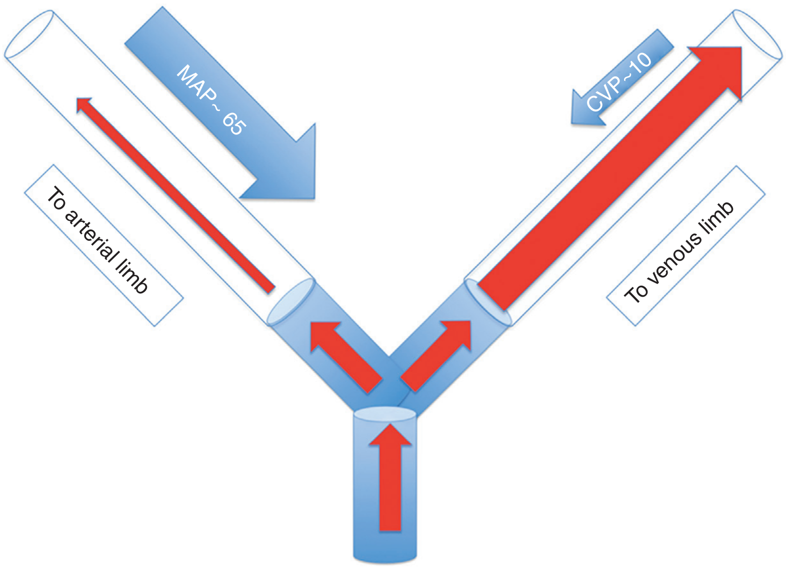

FIG. 14.14 Preferential flow of blood toward venous limb on VAV ECMO
CVP, Central venous pressure; MAP, mean arterial pressure.

HOW DO YOU KNOW HOW BLOOD IS BEING DIVIDED?

After going on VAV ECMO, you can attach a flow probe to the arterial limb. This will measure how much blood is going into the arterial side of the circulation (say 2 L/min). This can be subtracted from the total blood flow to give the amount of blood going to the venous cannula. So, if total blood flow is 5 L/min, you know you are giving 2 L/min to the arterial limb and 3 L/min to the venous limb.

If you want to then increase the arterial blood flow to achieve 2.5 L/min of flow, you can attempt to tighten the clamp on the venous limb, increasing the resistance, and directing more blood to the arterial side. However, at a certain point, you may reach diminishing returns, as tightening the occlusion to the venous limb may increase the overall resistance applied to the circuit and decrease the overall blood flow.

PUTTING IT ALL TOGETHER

Understanding retrograde physiology is essential to understanding peripherally inserted VA ECMO support. Knowing the impact of flow of this nature and being able to identify and predict the risks associated with this flow is what defines candidacy for VA ECMO as opposed to other forms of mechanical circulatory support. The degree that a patient with cardiac failure will be able to tolerate this retrograde flow is the fundamental component of whether or not they should be supported with VA ECMO.

This dynamic changes slightly when we think of the physiology of centrally cannulated ECMO, which is what we will focus on in the next chapter.

SUGGESTED READING

Avalli, L., Maggioni, E., Sangalli, F., Favini, G., Formica, F., Fumagalli, R. et al. (2011). Percutaneous left-heart decompression during extracorporeal membrane oxygenation: an alternative to surgical and transeptal venting in adult patients. *Asaio Journal*, *57*(1), 38–40.

Baldetti, L., Gramegna, M., Beneduce, A., Melillo, F., Moroni, F., Calvo, F., et al. (2020). Strategies of left ventricular unloading during VA-ECMO support: a network meta-analysis.. *International Journal of Cardiology*, *312*, 16–21.

Funk, D. J., Jacobsohn, E., & Kumar, A. (2013). The role of venous return in critical illness and shock—part I: physiology. *Critical Care Medicine*, *41*(1), 255–262.

Fux, T., Holm, M., Corbascio, M., van der Linden, J., Lund, L. H. et al. (2017). Pre-implant outcome predictors in patients with refractory cardiogenic shock supported with VA-ECMO. Journal of the American College of Cardiology, 70(16), 2094–2096.

Tarazi, R. C., & Levy, M. N. (1982). Cardiac responses to increased afterload. State-of-the-art review. *Hypertension*, *4*(3_pt_2), 8–18.

Zweck, E., Thayer, K. L., Helgestad, O. K. L., Kanwar, M., Ayouty, M., Garan, A. R., et al. (2021). Phenotyping cardiogenic shock. Journal of the American Heart Association, 10(14), e020085.